



Alberta Public Health Disease Management Guidelines

Dengue

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Health and Wellness Promotion Branch

Public Health Division

Alberta Health

DENGUE | Alberta Health, Government of Alberta

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Case Definition

Confirmed Case

Laboratory confirmation of infection with clinical illness^(A):

- Detection of Dengue virus (DENV) nucleic acid (e.g., by RT-PCR) or isolation from an appropriate clinical specimen (e.g., blood, CSF, urine);

OR

- Positive DENV IgM antibody with DENV IgG seroconversion and the following were not completed: ZIKV IgM/IgG and PRNT testing; ^(B)

OR

- Positive DENV IgM antibody with or without DENV IgG seroconversion AND a positive ($\geq 1:20$) DENV Plaque Reduction Neutralization test (PRNT) where the ZIKV PRNT is negative; ^(B)

Probable Case

Laboratory evidence of infection with clinical illness^(A):

- Positive DENV IgM antibody with or without positive DENV IgG antibody and the following were not completed: ZIKV IgM/IgG and PRNT testing. ^(B)

(Refer to Table 1 for more information).

^(A) Clinical illness: refer to [Clinical Presentation](#).

^(B) Due to declining ZIKV activity worldwide, PRNT testing for flaviviruses such as dengue and zika will only be requested when ZIKV is suspected (usually in congenital cases). Refer to Table 1 for assistance with interpreting lab results and/or consult the Virologist-on-call.

Table 1: Zika and Dengue Lab Result Interpretation⁽¹⁾*(Highlighted light blue are the most important when reviewing results)*

NOTIFIABLE							
NAT/PCR Testing	Zika Testing			Dengue Testing			Interpretation
	ZIKV IgM	ZIKV IgG	ZIKV PRNT	DENV IgM	DENV IgG	DENV PRNT	
ZIKV +	Any result ^(C)	Any result	Any result	Any result	Any result	Any result	CONFIRMED ZIKA
Negative Not tested	Positive	Any result	≥20 (Positive)	Any result	Any result	Negative	CONFIRMED ZIKA (dengue – not a case)
Negative Not tested	Positive	Any result	Not tested	Negative, Not tested	Negative, Not tested	Not tested	PROBABLE ZIKA
DENV +	Any result	Any result	Any result	Any result	Any result	Any result	CONFIRMED DENGUE
Negative Not tested	Not tested	Not tested	Not tested	Positive	Sero-conversion	Not tested	CONFIRMED DENGUE
Negative Not tested	Any result	Negative, Not tested	Negative^(D)	Positive	Any result	≥20 (Positive)	CONFIRMED DENGUE (zika – not a case)
Negative Not tested	Negative, Not tested	Negative, Not tested	Not tested	Positive	Any result	Not tested	PROBABLE DENGUE However, if subsequent testing show the following results, this indicates NOT A CASE : e.g., DENV IgM positive/IgG negative (no seroconversion) OR DENV IgM negative /IgG negative (false positive IgM).
Negative Not tested	Any result	Any result	≥20 (Positive)	Any result	Any result	≥20 (Positive)	PROBABLE FLAVIVIRUS – specific virus cannot be identified due to cross-reacting flavivirus antibodies. Refer to Flavivirus case definition.

NOT NOTIFIABLE – For information only to assist with case management and classification						
ZIKV IgM	ZIKV IgG	ZIKV	DENV IgM	DENV	DENV PRNT ^(F)	Interpretation
Pos, Neg ^(F) or Equiv ^(E)	Pos or Neg	Pending	Pos, Neg or Equiv	Any result	Pending	SUSPECT CASE: zika or dengue, pending lab results
Any result	Pos or Neg	Neg	Pos, Neg or Equiv	Any result	Neg	NOT A CASE: No evidence of zika or dengue, however if a specimen was collected <5 days after onset the PRNT results may be negative. Convalescent specimens are recommended in 14 days.
Equiv, Neg or Not tested	Pos or Neg	≥20 (Positive)	Neg	Any result	Neg	NOT A CASE: Evidence of zika infection; likely remote infection > 3months (Not an acute case)
Equiv, Neg or Not tested	Pos or Neg	Negative	Equiv, Neg or Not tested	Any result	≥20 (Positive)	NOT A CASE: Evidence of dengue infection; likely remote infection > 3months (Not an acute case)
Not tested			Neg or Equiv ^(E)	Any result	Not tested	NOT A CASE: Equivocal, indeterminate or inconclusive results indicate that result could not be interpreted as positive or negative. New samples should be collected if clinically indicated

^(C) “Any result” includes “Not tested”. Applicable where ever this phrase is used in the table.^(D) For the purposes of lab result interpretation, a negative PRNT is any value less than 20.^(E) Equivocal, indeterminate and inconclusive results are indicative of laboratory samples that could not be interpreted for various reasons (e.g., borderline positive or non-specific binding) in order to give a result of positive or negative. If clinically indicated, new specimens should be collected for these individuals. Applicable where ever this phrase is used in the table.^(F) Individuals tested >3 months after exposure will almost always test IgM negative. If clinically indicated (e.g., pregnancy) samples may be forwarded to NML for PRNT testing with incomplete serology.

Reporting Requirements

Physicians, Health Practitioners and Others

- Physicians, health practitioners or others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

- Laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the MOH (or designate) of the zone and the Chief Medical Officer of Health (CMOH) (or designate).

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- The MOH/designate of the zone where the case currently resides shall report all confirmed and probable cases to the CMOH/designate using the *Notifiable Disease Report (NDR)* form or another mutually agreed upon reporting system within two weeks of notification for the initial NDR and the final NDR (amendment) within four weeks, as per usual.
- For out-of-province and out-of-country cases, the following information should be forwarded to the CMOH/or designate as per usual communicable disease reporting processes and include:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - any positive laboratory report/s, and
 - other relevant clinical / epidemiological information.

Epidemiology

Etiology

DENV is a member of the *Flaviviridae* family.^(2,3) There are four closely related but antigenically distinct virus serotypes: DEN-1, DEN-2, DEN-3, and DEN-4.

Clinical Presentation

Dengue infection is a systemic and dynamic disease. It has a wide clinical spectrum which includes both severe and non-severe clinical manifestations. The majority of individuals have no symptoms, especially children and those who have not had DENV previously.⁽⁴⁾ Illness begins abruptly and has three phases -- febrile, critical and recovery.

Dengue: Usually lasts 2-7 days;⁽⁵⁾

Definition: fever and any two of the following:

- nausea/vomiting,
- rash,
- myalgia, arthralgia,
- leukopenia, and
- minor hemorrhagic manifestations (e.g., petechiae, ecchymosis, purpura, epistaxis, bleeding gums, hematuria, or a positive tourniquet test).

Dengue with Warning Signs:** progression to severe dengue begins to occur in the late febrile phase around the time of defervescence and includes any of the following:⁽⁵⁾

Definition: Dengue with any of the following

- abdominal pain or tenderness,
- persistent vomiting,
- Clinical fluid accumulation (ascites, pleural effusion),
- Mucosal bleeding,
- Lethargy, restlessness,
- Liver enlargement >2 cm, and
- Laboratory: increase in hematocrit concurrent with rapid decrease in platelet count.

***requires strict observation and medical intervention*

Severe Dengue: In the critical phase individuals with severe dengue develop severe plasma leakage as a result of a marked increase in vascular permeability. Intense medical intervention is required to aid their physiologic compensatory mechanisms in maintaining adequate circulation. However, once hypotension develops, systolic blood pressure rapidly declines, and irreversible shock and death may ensue despite resuscitation.

Definition: Dengue with at least one of the following criteria (includes Dengue Hemorrhagic Fever & Dengue Shock Syndrome [DSS]):⁽⁵⁾

- Severe Plasma Leakage leading to:
 - DSS
 - Fluid accumulation with respiratory distress
- Severe Bleeding as evaluated by clinician
- Severe organ involvement
 - Liver: AST or ALT $\geq$ 1000
 - CNS: impaired consciousness
 - Failure of heart and other organs

Mortality rates from severe dengue can be as high as 20% if untreated.⁽⁶⁾ Pregnant women with DENV are at increased risk of miscarriage, premature birth and infants with low birth weight.⁽⁷⁾

Diagnosis

PCR testing by Alberta Precision Laboratories (APL) should be performed on plasma (blood in EDTA) serum collected during the first two weeks after symptom onset.⁽⁹⁾ DENV IgM antibodies are usually detectable 6 – 7 days after the onset of illness and generally indicate a current or recent infection. A serum specimen should be collected for acute phase DENV IgM antibody testing at least 5 days after onset of clinical illness. A convalescent phase serum should be obtained at least 2 weeks after the first sample has been collected.

Antibody detection should be performed on whole blood in SST tubes.⁽⁹⁾ Currently, the most frequently acquired flavivirus in returning travelers is dengue. However, flavivirus antibodies can cross-react in the existing serological tests, especially those for IgM, resulting in some uncertainty in IgM results between ZIKV and DENV. If ZIKV is suspected, cases with positive or equivocal serology are sent to the National Microbiology Laboratory (NML) in Winnipeg to be confirmed using PRNT. This is mainly indicated in scenarios where congenital ZIKV infection is suspected after consultation with the Virologist-on-call.

PRNT is not always able to provide a definitive determination of the specific flavivirus causing a recent infection, especially if the person has a prior history of flavivirus infection (e.g., previous DENV or West Nile virus), is immunized against a flavivirus (e.g., yellow fever or Japanese encephalitis) or is currently co-infected with another flavivirus. These cases may have test results such as DENV IgM negative with positive PRNT results or positive PRNT results for both DENV and ZIKV, making it difficult to determine which virus is causing the illness (i.e., probable flavivirus).

(Refer to the [ProvLab website](#) (specifically the Guide to Services) for the latest information regarding diagnosis and testing).

Treatment

- There is no effective, specific therapy for DENV.⁽³⁾
- Symptomatic care should be made available if the case develops dehydration or severe disease.⁽³⁾
- Acetylsalicylic acid (ASA) and other non-steroidal anti-inflammatory drugs (NSAIDs) are contraindicated as they may aggravate the bleeding tendency associated with some DENV infections.^(3,10)
- The risk of death can be reduced from 10% to 1% in severe cases if shock is recognized early and intensive supportive therapies are initiated quickly.⁽¹⁰⁾

Reservoir

DENV is primarily maintained in a human - *Aedes* mosquito cycle.^(3,11)

Populations of *Aedes (aegypti or albopictus)* mosquitoes are not established in Canada. The likelihood that *Aedes* mosquitoes will become established in Canada is considered very low as they are limited by the lack of environmental and climate suitability here.⁽¹²⁾

Transmission

DENV is mainly transmitted by the bite of the infected *Aedes* mosquitoes.⁽³⁾

Although very rare, transmission may occur via needlestick, blood transfusion (especially in endemic countries), organ transplant and vertically, especially if the mother is acutely ill around the time of delivery.^(2,3,11,13)

The virus is sensitive to heat and is susceptible to many common disinfectants including ethanol, sodium hypochlorite, and glutaraldehyde.⁽²⁾ It is, however, stable in dried blood and exudates for up to 9 weeks at room temperature.^(2,14)

Incubation Period

The incubation period ranges from 3 – 14 days, but is typically 4 – 7 days.⁽³⁾

Period of Communicability

DENV has a short period of communicability via blood, from shortly before until the end of the febrile period (usually 3 – 5 days).⁽³⁾

Host Susceptibility

Susceptibility is universal. Those living in or travelling to tropical areas are at highest risk of exposure to DENV. Those who have been previously infected, babies and young children are more likely to get severe infection.⁽³⁾

Infection and recovery from one serotype was once thought to provide lifelong immunity to that serotype but no other serotypes.⁽¹⁵⁾ Recent evidence suggests re-infection with the same serotype is possible however, more research is required.⁽¹⁶⁾ Evidence also suggests that a subsequent infection by a different serotype increases the risk of acquiring DHF.⁽¹¹⁾ Infants who were infected with DENV between 6 – 12 months of age, and born to previously DENV-infected mothers, are at increased risk for severe disease.⁽³⁾

Incidence

DENV is not nationally notifiable. It has been notifiable in Alberta since 1985.⁽¹⁷⁾ Between 2010 and 2015, less than 70 travel-acquired cases were reported to Alberta Health. DENV is not considered endemic in Alberta. For more information on the occurrence of dengue worldwide refer to the [CDC Yellow Book webpage on dengue](#).

Public Health Management

Key Investigation

- Confirm the diagnosis, if possible. Other arboviral diseases such as chikungunya, ZIKV or West Nile virus and rash illnesses such as measles, rubella, parvovirus, enterovirus, or febrile illnesses such as malaria should be ruled out.⁽³⁾ Co-infection with multiple pathogens in returning travelers can occur.
- Determine where the case may have been exposed, taking into consideration the incubation period, reservoir, and mode of transmission, including a history of:
 - recent travel to an endemic area,
 - recent emigration from an endemic area, or
 - History of recent blood transfusion (within previous 21 days).

Management of a Case

- The following information should be provided to cases depending on their needs:
 - DENV cases should defer donating blood for 56 days after the onset of symptoms; this may vary depending on the country visited and length of time spent there. Refer to [Canadian Blood Services](#) website for more information.
 - Provide education how to prevent [transmission](#) and infection (i.e., preventive measures below).
 - Contact health care provider if symptoms worsen (refer to [Clinical Presentation](#)).

Management of Contacts

- There is no direct follow-up of contacts; however, it may be prudent to identify others who may have been exposed to the same source as the case to find undiagnosed cases.

Preventative Measures

- Provide the following recommendations to persons travelling to endemic areas:
 - Consult a health care provider or visit a travel health clinic preferably six weeks before travel.
 - Protect yourself from mosquito bites:⁽¹⁸⁾
 - remain in well-screened or completely enclosed, air-conditioned areas whenever possible,
 - wear light-colored clothing and cover up as much as possible with long pants, long-sleeved loose fitting shirts, shoes/boots (not sandals) and a hat.
 - sleep under a bed net, preferably treated with insecticide, and
 - use insect repellent (containing DEET or Icaridin/picardin), 15 minutes after sunscreen, on exposed skin and on clothing.
 - There is currently no DENV vaccine licensed for use in Canada.

ANNEX A: Revision History

Revision Date	Document Section	Description of Revision
August 2021	Case Definition	• Addition of IgG seroconversion to confirmed case definition • Footnote added regarding change to dengue PRNT testing
	Lab interpretation Table	• Updated to reflect change to case definition
	Reporting Requirements	• Removed "in the prescribed form" from physicians, health practitioners and others section.
	Diagnosis	• Moved to Epidemiology section and updated to reflect changes in testing
	Treatment	• Moved to Epidemiology section
	Management of a Case	• Added education and what to do if symptoms worsen
	General	• New Template • Refreshed web links
August 2022	Table 1	• Added equivocal result to Dengue testing as per new testing protocol at ProvLab.

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